Transitive label budgets on six residual fixed points

Human Compression project: L044 continuation

Use the critical singleton-end setting of the accepted residual-deficit proof: $G = \langle \rho_0, \ldots, \rho_{r-1} \rangle \leq A_n$ is transitive, is a connected string C-group, and has every maximal omission intransitive; G_0 has orbits $\{a\}$ and its complement; $\rho_0 = (a\ b)\delta$, $K = G_{>1}$, $R = G_{>2}$ and $\sigma = \rho_1$. The odd involution δ has t pairs, with union P, and an even fixed set F of size f in $\Omega \setminus \{a, b\}$. Assume $n = 2r \geq 14$, so K has $d = t + f/2 - 1$ labels. All block-image omission-orbit statements below use the complete accepted actual-deficit proof. Whiston's bound and the complete odd-degree maximal-string theorem, including degree five, remain paid.

Lemma 1 (Single-transposition restrictions)**.** *If K is transitive on F, $f \geq 4$, and all nonidentity distinguished restrictions to F are single transpositions, they are exactly the $f - 1$ successive edges of a natural path, each with one label.*

Proof. Their edge graph is connected. Three distinct edges incident to one vertex would have pairwise noncommuting labels, impossible in a string. A cycle is likewise impossible: labels chosen on its distinct edges would require its line graph, a cycle, to embed in the path of label positions. Hence the simple edge graph is a path.

Choose a label on each edge of that path. Their positions form a path in the integer line, so are consecutive in one of the two orders. Any repeated label on an interior edge would have to be adjacent to both chosen neighboring labels, whose unique common neighbor is the already occupied position. For an end edge, its neighboring edge is interior because $f \geq 4$; both positions adjacent to that neighboring label are already occupied. A second end-edge label is also impossible. □

Lemma 2 (Minimum transitive fixed-set budget)**.** *Let E be a set of h distinguished K labels whose restrictions are transitive on F. Then $h \geq f/2$. If equality holds and $t \geq 3$ is odd, the other $t-1$ pair images generate the ordinary S_t path, at consecutive positions J, and every label of E has trivial pair image.*

Proof. Deleting a label outside E leaves the fixed-set action transitive. Its pair image cannot be generated by the other pair images, as that would leave the full block-orbit partition unchanged. In particular the $d - h$ pair images outside E are independent. Whiston gives $d - h \leq t - 1$, or $h \geq f/2$.

At equality those $t - 1$ images generate S_t and form its ordinary path by the accepted odd-degree maximal-string theorem. For $t = 3$ this just says that two independent involutions in S_3 are distinct transpositions. Consecutive noncommutations force consecutive positions in the original string. Omitting any path mark leaves two pair orbits. If an E label joined them, the full omission would be transitive both on pairs and on F, contradicting the actual orbit deficit. Thus every E label preserves every path cut; their common stabilizer fixes every pair-block. This proves the last assertion. □

Theorem 3 (The three-label six-point branch)**.** *Suppose $f = 6$ and three distinguished K restrictions are transitive on F. Then necessarily $t = 3$, $n = 14$, $r = 7$, and ρ_2 is one of those three labels. In particular the case $t \geq 5$ is impossible.*

Proof. Let E be the three labels. By Lemma 2 their pair images are trivial, while J is an ordinary path on the t pairs. Here $t \geq 3$ because $n = 2t + 8 \geq 14$.

First suppose $\rho_2 \in J$. Then J starts at index2 and all three extras follow it. The group R retains E, so F is an R-orbit of size six. On pairs, R fixes the first endpoint pair and is transitive on the other $t-1$. Its point orbits in P therefore have sizes at most two, $t-1$, or $2(t-1)$: in a transitive pair-block orbit, a point orbit meets every block in the same number of points. The only possible size-six collision for odd t is $t = 7$, with a row meeting each of the six pair-blocks once. Every E label has trivial pair image and preserves that R-orbit, so fixes it pointwise. It cannot be $\sigma(F)$, since equivariance would then make all of E fix F, whereas E is transitive on F. In every case σ must preserve F. So does K, contradicting G_0 transitivity.

Thus $\rho_2 \in E$. Now R contains all of J, so its point orbits in P have size t or $2t$. If $t \geq 7$, their sizes exceed six, and P is their invariant union under σ. This again contradicts G_0 transitivity. If $t = 5$, the only possible collision is a five-point row $O \subseteq P$ and an R-orbit $U \subseteq F$ of size five. The two surviving E labels have trivial pair images, and hence fix O pointwise. If σ exchanged O, U, they would fix U pointwise as well, and therefore also fix its single-point complement in F. Then the entire E action on F would be generated by the one omitted involution, which is not transitive on six points. So that collision cannot occur; σ preserves P, again impossible. Only $t = 3$ remains. □

Literal six-point restrictions. For a transitive action on six fixed points, a distinguished triple transposition supplies three disjoint edges. Extend them to a spanning tree of the full fixed-set edge graph; just two further edges are needed, so at most three distinguished labels are transitive on F. Lemma 2 makes the minimum exactly three, and Theorem 3 applies. A double transposition similarly gives a transitive set of at most four labels. If all restrictions are single transpositions, Lemma 1 puts the action within the separately proved natural six-point-path exclusion.

Attribution and boundary. The source is Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, physical pages38–44. The path-label observation extends the separate accepted four-fixed proof by `rewrite_termination`; the transitive-label budget and the present six-point reductions use that proof's actual-omission and equivariant-orbit approach. The exact natural six-point exclusion is a separate new bounded dependency, not part of Lemma 1 alone.

The $n = 14$ three-label boundary is not excluded here. Nor is the four-label six-point case, an intransitive action on six fixed points, or any general larger fixed set. The theorem performs no finite enumeration and invokes no projected IP, alternating induction, whole-source recovery, new original credit or conjecture credit. Distinct review remains pending.